Last name _____

First name _____

LARSON—MATH 310–CLASSROOM WORKSHEET 22
Matrix Inverses and Pseudo-Inverses.

Review

1. Suppose Q is an orthogonal matrix. What is Q invertible? What is Q^{-1}?
2. Why is any lower-triangular matrix L with **non-zero** entries on the diagonal invertible?
3. Why is any upper-triangular matrix U with **non-zero** entries on the diagonal invertible?
4. How can we *find* the inverse of a upper-triangular matrix R with **non-zero** entries on the diagonal?
5. How can we use Gram-Schmidt to find the inverse of a square matrix (with linearly independent columns)?
6. What is *back substitution*? If R is an upper-triangular matrix with non-zero entries on the diagonal, how can we solve any system $R\vec{x} = \vec{b}$?

Matrix Inverses!

Let

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$$

1. We know $A = QR$ for an orthogonal matrix Q and an upper-triangular matrix R. Find Q.

2. We know orthogonal matrices Q are invertible with inverse $Q^{-1} = Q^T$. So, $Q^T A = R$. Use this fact to find R (In particular $R_{i,j} = (Q^T A)_{i,j}$. So $R_{i,j}$ is the inner product of $\vec{q}_i$ and $\vec{a}_j$).

3. Check that A has linearly independent columns (for a square matrix that's enough to guarantee that A has an inverse A^{-1}).

4. We know $A^{-1} = (QR)^{-1} = R^{-1}Q^{-1} = R^{-1}Q^T$. So to find A^{-1} we just need R^{-1}. Since R is upper-triangular with no zero entry on its diagonal we know it has an inverse. Solve

$$R\left[\vec{x}_1\vec{x}_2\right] = \left[\vec{e}_1\vec{e}_2\right],$$

by solving $R\vec{x}_1 = \vec{e}_1$ and $R\vec{x}_2 = \vec{e}_2$. (These correspond to systems that can be backsolved.)

5. Let $R^{-1} = \left[\vec{x}_1\vec{x}_2\right]$. Check that $RR^{-1} = I$.

6. Now find A^{-1} (we know $A^{-1} = R^{-1}Q^{-1} = R^{-1}Q^T$).

7. Check that $AA^{-1} = I$.